

Satellite based airglow/aurora observations during the April 2023 geomagnetic storm



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1. Background

Solar activity, including variations in solar radiation and transient disturbances in the solar wind, drives a variety of processes in Earth's atmosphere. Solar ultraviolet (UV) radiation provides the primary energy input to the mesosphere and lower thermosphere, while enhanced solar wind plasma and magnetic field variations can indirectly influence atmospheric dynamics through magnetospheric coupling. These processes lead to atmospheric emissions observed as airglow and, at higher latitudes, aurorae.

MATS (Mesospheric Airglow/Aerosol Tomography and Spectroscopy) satellite images provide an opportunity to study the variability of O₂ A-Band airglow in relation to solar UV flux and solar wind.

The geomagnetic storm on April 23rd-24th 2023 was the first major storm of the current solar cycle. The configuration of the magnetosphere was different from typical conditions because of an interval of sub-Alfvénic solar wind.

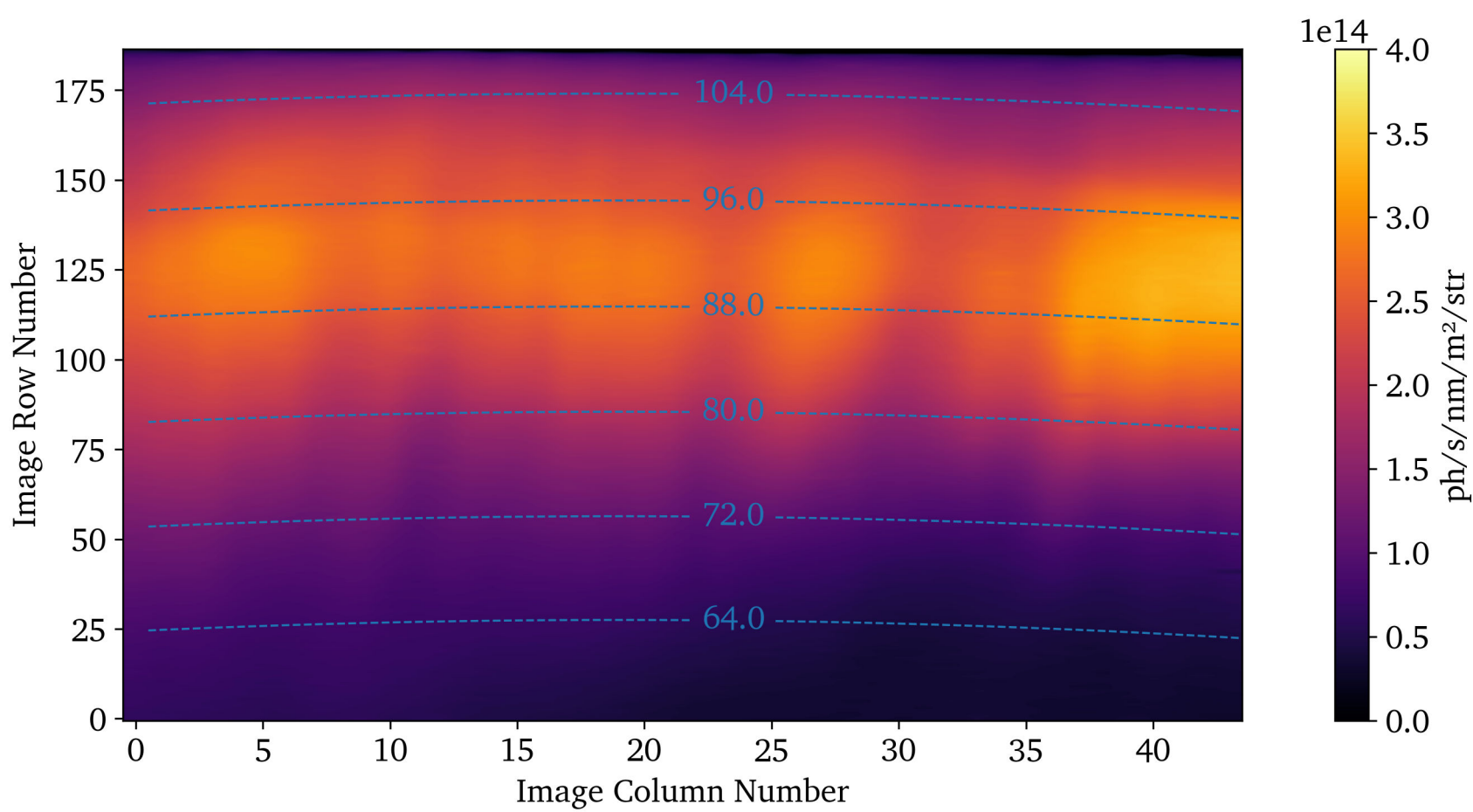


Figure 1: Limb observation of aurora emissions taken by MATS IR2 channel after the arrival of the magnetic ejecta on 2023-04-24T05:17:14. Intensity in units of photons/s/nm/m²/steradians. Altitude at tangent point in km as blue dashed lines. Images are taken every 6 seconds

2. Data

To investigate the presence of correlations between airglow/aurora intensity as measured by MATS and sun driven disturbances, this study examines data from the following sources:

- MATS IR2 images [1]
- UV and X-Ray flux from SDO/EVE [2]
- Solar Wind parameters from OMNI [3]
- Coronal Mass Ejection (CME) event list [4]
- Streaming Interaction Region (SIR) event list [5]

3. Methods

Using the available MATS observations from February to May 2023, we extracted a time series of airglow brightness variations in the mesosphere by taking the mean of each image. Dayside and nightside sections of the orbit are classified according to solar zenith angle at the tangent point. Difference from average brightness at each solar zenith angle is calculated.

4. Results

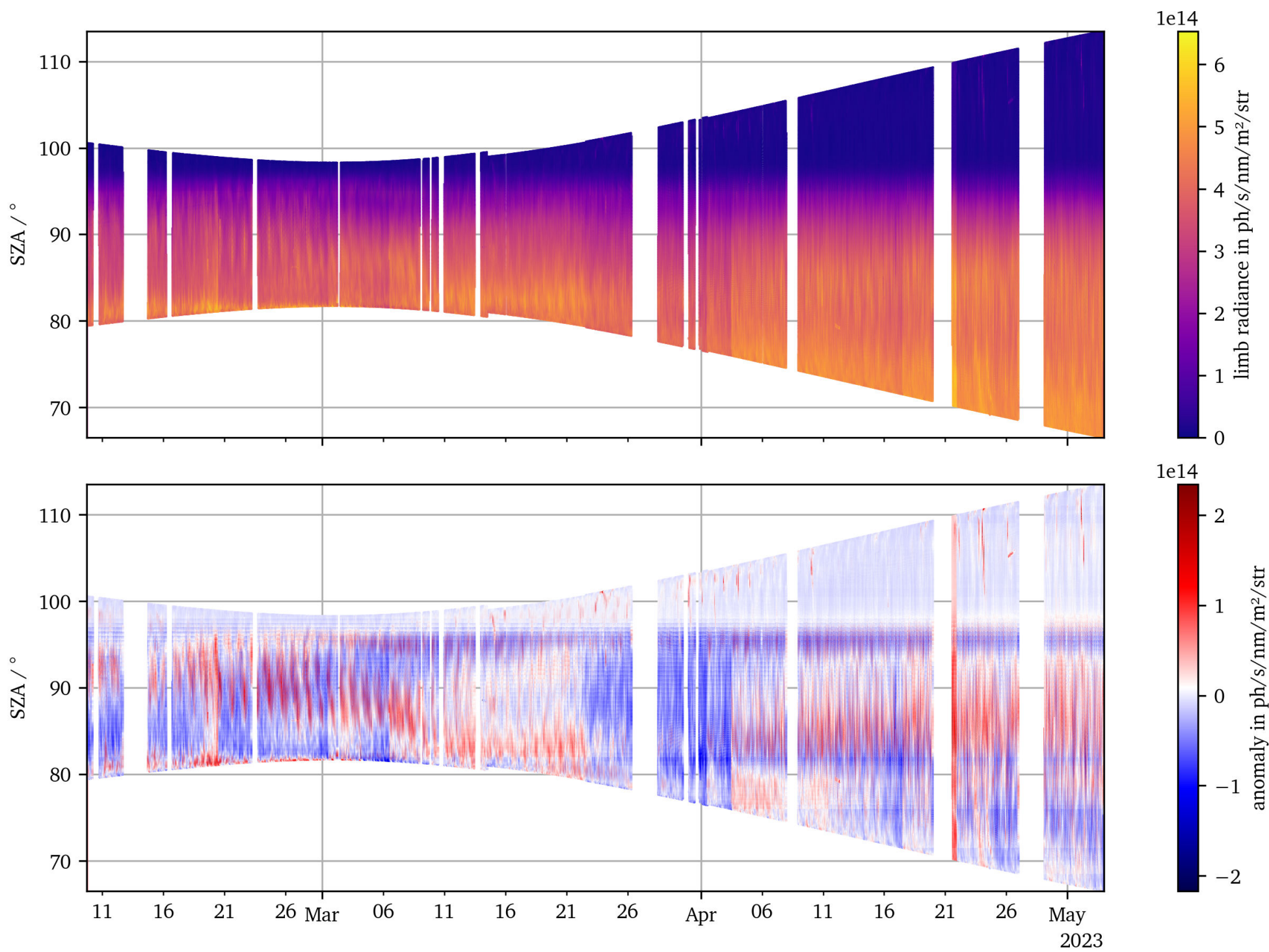


Figure 2: Overview of airglow from February to May 2023. Top panel: Airglow emission strength as a colormap over solar zenith angle (SZA) Bottom panel: deviation from average brightness at each SZA

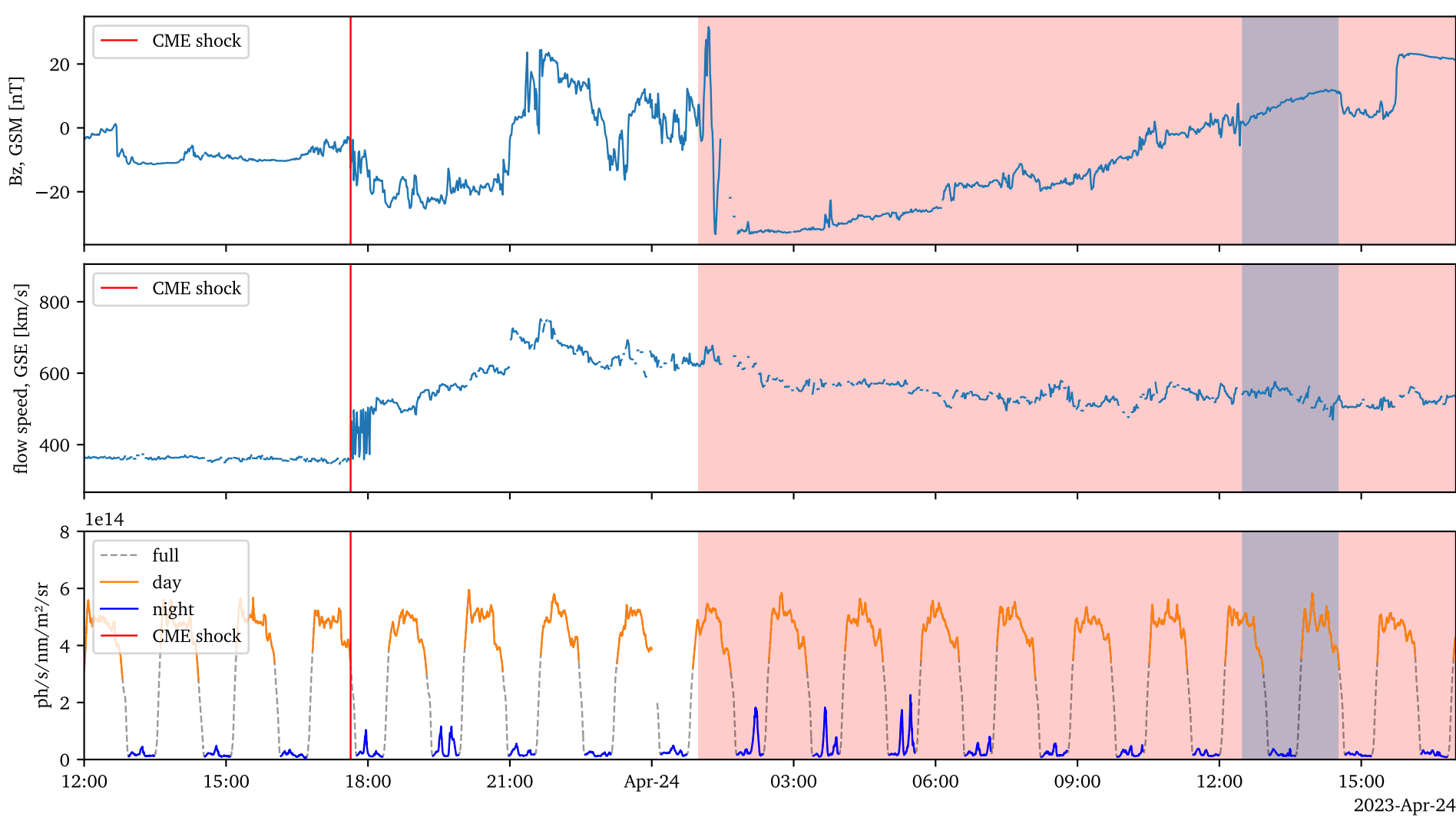


Figure 3: Solar wind conditions and airglow/aurora measurements during the April 2023 storm. IMF B_z component in nT, solar wind velocity in km/s, MATS limb radiance dayside (orange) nightside (blue). CME shock arrival (red horizontal line), CME ejecta (shaded red box), subalfvénic solar wind (shaded blue box)

5. Conclusion

Airglow: Long term trends seem to be mainly dependent on SZA rather than changes in EUV flux. Atmospheric phenomena such as planetary waves and tides cause greater disturbances than X-class solar flares.

Aurora: The most intense emissions (5-10x greater than nightglow baseline) occur during times of negative B_z . Aurora activity during the later phases of the storm (including the sub-Alfvénic interval) is low.

Bibliography

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